

Reasoning Patterns for Efficient AI Coding: An Empirical Study

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Long reasoning traces generated by Large Reasoning Models (LRMs) provide a rich behavioral record for understanding and controlling model inference, yet the underlying reasoning patterns and how different patterns affect coding models remain poorly understood, limiting systematic characterization and effective utilization of model reasoning behavior.

We conduct an empirical study of LRM’s reasoning patterns in coding tasks. We transform reasoning traces into structured action sequences and mine reusable patterns[**gu: how do we mine patterns?**]. Our findings characterize [**gu: XXX**] general reasoning patterns: for example, code generation follows a design-to-implementation paradigm, whereas debugging follows an evidence–fix–recheck loop. We further investigate fine-grained patterns that distinguish positive and negative reasoning traces. We find that positive reasoning traces demonstrate a clear pattern of verification-driven convergence, while negative reasoning traces are characterized by repeated design revision, reflection, and knowledge recall without closure.

As prototype-of-application, we utilize the identified patterns for inference-time applications across the evaluated task–model settings[**gu: ??**]. Our results show that injecting general patterns into model’s reasoning traces enables effective task identification from reasoning traces[**gu: not clear**]. While positive patterns improve overall accuracy of downstream tasks, early stopping negative patterns substantially reduces API-reported token usage with only a modest decrease in pass@1. All applications also generalize to previously unseen models. These results demonstrate that reasoning patterns can support both the analysis and inference-time control of LRM behavior in AI coding.

Additional Key Words and Phrases: reasoning patterns, AI coding, reasoning trace analysis, reasoning pattern mining, inference-time optimization

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1 Introduction

Large reasoning models (LRMs) are rapidly reshaping AI coding [35, 36]. Unlike conventional code language models that generate an answer directly, LRMs produce a sequence of intermediate reasoning steps before arriving at a solution. This capability is particularly valuable for coding tasks, where solving a problem often involves intermediate decisions, such as decomposing requirements and designing an algorithm for code generation or tracking program states during execution reasoning. The reasoning process is therefore exposed as part of the model’s output rather than remaining implicit.

The availability of these reasoning traces has led to a growing body of research on their role in code-related tasks [6, 29]. Existing studies show that intermediate reasoning can improve task

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performance, but also reveal substantial inefficiencies. Reasoning traces may contain incorrect assumptions, redundant exploration, repeated revisions, and prolonged deliberation that adds inference cost without improving the final result [4, 12, 18, 39]. These observations have motivated the use of reasoning traces for purposes beyond generating an answer, including self-correction [20, 22], evaluation [26, 30], and inference-time control [9, 10].

However, most existing studies examine reasoning from the perspective of individual tasks or specific behaviors. They focus on final task performance [3, 15, 16], prompting strategies [13, 27, 29], or particular reasoning behaviors in tasks such as code generation [12, 18, 31, 39]. This leaves a more basic question largely unanswered: do reasoning traces contain recurring structures that characterize how LRMs solve coding tasks? If such structures exist, it is also unclear which of them are associated with successful or unsuccessful reasoning, and whether they can be used to improve the efficiency and reliability of inference. Addressing these questions requires looking beyond individual reasoning traces and examining them as sequences of recurring reasoning actions.

To address these questions, we conduct an empirical study of reasoning patterns in AI coding. We collect reasoning traces generated by LRMs across four representative coding tasks: code generation, code execution reasoning, program debugging, and code translation. We transform these traces into structured reasoning action sequences using a unified action taxonomy. This representation allows us to analyze reasoning in terms of observable actions rather than the surface form of generated text. We then examine reasoning patterns at two levels. Macro patterns capture task-level reasoning structures, while micro patterns characterize fine-grained behaviors within individual reasoning trajectories. Finally, we investigate whether these patterns can support practical applications for analyzing and controlling inference.

Specifically, we investigate the following research questions:

- **RQ1 (General Patterns): What task-level reasoning patterns emerge across different coding tasks?**

Different coding tasks impose distinct reasoning demands, yet it remains unclear whether these differences are reflected in recurring task-level structures. To characterize such structures, we transform reasoning traces into action sequences using a unified action taxonomy and mine recurring action fragments across the four coding tasks. We then examine the resulting patterns to identify the reasoning strategies that characterize each task.

- **RQ2 (Positive and Negative Patterns): What fine-grained reasoning patterns distinguish valid from invalid reasoning trajectories?**

Task-level patterns capture differences across coding tasks, but they do not explain why individual reasoning trajectories succeed or fail. We therefore compare valid and invalid reasoning traces and mine recurring action fragments that distinguish the two groups. For tasks with objective correctness signals, trace validity is assessed using an ensemble LLM-as-a-judge protocol. We then identify both positive patterns associated with valid reasoning and anti-patterns associated with invalid reasoning.

- **RQ3 (Applications): Can the discovered reasoning patterns be leveraged to improve AI coding performance?**

We further examine whether the discovered patterns can be used to guide and control inference. Specifically, we evaluate three applications. First, we use macro-pattern features to identify coding tasks from reasoning traces. Second, we augment prompts with micro-pattern guidance to influence reasoning behavior. Third, we use pattern-aware early stopping to terminate inference based on detected reasoning patterns and measure its effect on coding accuracy and token usage.

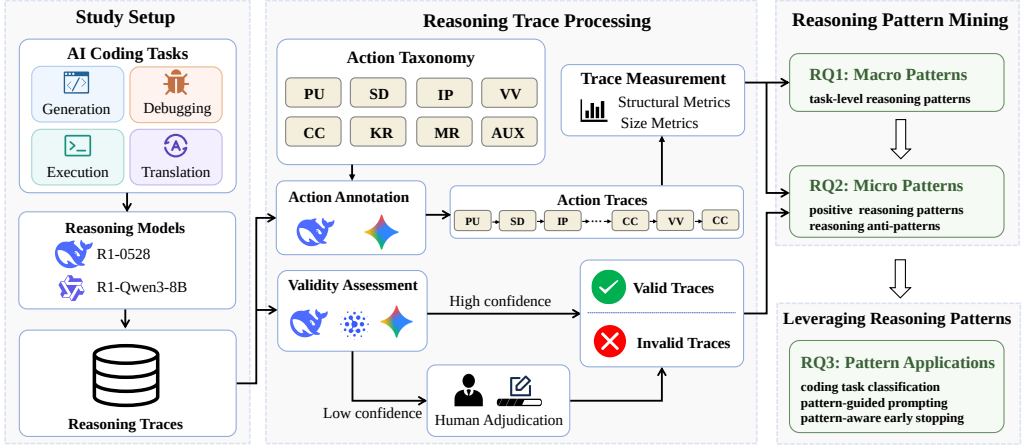


Fig. 1. Overview of the Empirical Study.

Our study yields three main findings. Coding tasks exhibit distinct macro-level reasoning structures: code generation follows a design-to-implementation flow, code execution reasoning centers on state tracing, debugging follows an evidence-fix-verification loop, and code translation centers on semantic mapping. At the micro level, valid and invalid trajectories differ systematically. Valid traces exhibit verification-driven convergence, whereas invalid traces are characterized by recurring behaviors such as repeated redesign, reflection, and knowledge recall without resolution. Beyond characterization, the discovered patterns also support practical applications across the evaluated task-model settings. Macro patterns support task identification, micro-pattern-guided prompting improves single-shot correctness, and pattern-aware early stopping substantially reduces API-reported token usage while incurring only a modest decrease in pass@1. [The three applications also generalize to previously unseen models.](#)

In summary, this paper makes the following contributions:

- We establish a unified reasoning-action representation for cross-task analysis of LRM reasoning traces and conduct an empirical study across four representative AI coding tasks.
- We characterize reasoning in AI coding at two levels, identifying task-specific macro patterns and fine-grained micro patterns that distinguish valid from invalid reasoning behaviors.
- We demonstrate the utility of these patterns for AI coding through three inference-time applications: task identification, pattern-guided prompting, and pattern-aware early stopping.

2 Methodology

Figure 1 presents an overview of our empirical study. First, we collect reasoning traces generated by two LRMs across four representative AI coding tasks and assess the validity of each trace. Second, we convert each trace into a sequence of reasoning actions based on a unified reasoning-action taxonomy and quantify the resulting behaviors using action- and trace-level metrics. Third, we mine recurring patterns from the resulting action traces, including task-level macro patterns for RQ1 and validity-related micro patterns and anti-patterns for RQ2. Finally, we evaluate how the

discovered patterns can be used for AI coding in RQ3, through coding task identification, pattern-guided prompting, and pattern-aware early stopping.

2.1 Study Setup

This section describes the coding tasks, datasets, and LRMs used in our empirical study.

2.1.1 Tasks and Datasets. To study reasoning patterns in AI coding across diverse problem-solving settings, we consider four representative coding tasks covering program synthesis, comprehension, maintenance, and transformation:

- *Code generation* requires a model to synthesize executable programs from natural-language specifications. We use LiveCodeBench v4–v6 [15], comprising 442 instances.
- *Code execution reasoning* requires a model to infer the output of a program through static reasoning rather than actual execution. We use the input–output prediction subset of CodeSense [21], comprising 382 instances.
- *Program debugging* requires a model to localize and repair defects in buggy programs. We use DebugBench [25], comprising 500 instances.
- *Code translation* requires a model to transform programs across programming languages while preserving their semantics. We use ClassEval-T [32], comprising 376 instances.

All four tasks provide objective correctness oracles, which allow us to analyze both reasoning traces and task outcomes under a unified experimental setting.

2.1.2 Models. For primary pattern discovery, we use two publicly available LRMs, DeepSeek-R1-0528 (denoted as R1-0528) and DeepSeek-R1-0528-Qwen3-8B (denoted as R1-Qwen3-8B) [6]. R1-Qwen3-8B is obtained by distilling R1-0528 into a Qwen3-8B backbone. For every task instance, both models are evaluated using identical prompts and decoding configurations. We collect the complete reasoning trace and final response produced by each model, resulting in the dataset used for subsequent trace processing and pattern analysis.

For pattern utilization, ****. [shen: complete this para.]

2.2 Reasoning Trace Processing

This section describes how we process the collected reasoning traces into reasoning action traces and the derived metrics used for subsequent reasoning pattern mining and analysis.

2.2.1 Reasoning Trace Collection. For each task instance, we instantiate the benchmark-provided prompt template with the corresponding instance-specific input, such as a problem description, starter code, buggy program, or source-language snippet. We submit each instantiated prompt independently to both models in thinking mode under a unified decoding configuration (temperature = 0.6, top- p = 0.95).

We extract the content enclosed by the <think> tags as the reasoning trace. To ensure completeness, we regenerate any response with an empty reasoning trace or lacking the information required for annotation or task evaluation, until a complete trace and task output are obtained.

After excluding incomplete, unparsable, or unlabeled responses, the resulting discovery dataset contains 3,376 reasoning traces across the four tasks and two models. Specifically, we retain 442/440 traces for code generation, 377/378 for code execution reasoning, 499/496 for program debugging, and 372/372 for code translation on R1-0528/R1-Qwen3-8B, respectively.

2.2.2 Reasoning Trace Validity Assessment. To distinguish valid from invalid reasoning behaviors in RQ2, we assess the validity of each reasoning trace using an ensemble LLM-as-a-judge

Table 1. Taxonomy of Reasoning Actions

Abbr.	Reasoning Action	Description
PU	Problem understanding	Understanding and clarifying the problem requirements.
SD	Solution design	Designing the high-level solution approach and strategy.
IP	Implementation planning	Planning the detailed implementation steps and logic.
CC	Code construction	Writing and refining the actual code.
VV	Verification and validation	Testing, debugging, and validating the solution.
KR	Knowledge retrieval	Retrieving and applying relevant knowledge.
MR	Meta-reasoning	Reflecting on and monitoring the reasoning process itself.
AUX	Auxiliary	Auxiliary statements that support the reasoning flow.

protocol [17, 40]. Trace-level validity is distinct from task-level functional correctness: a trace may lead to a correct final answer while containing logical inconsistencies, unsupported transitions, or unnecessary reasoning detours. We consider a reasoning trace *valid* if its reasoning is internally consistent and progresses coherently toward the final solution.

We employ three judge models, DeepSeek-V3.2 [7], Gemini-3-Flash [11], and GLM-5.1 [37]. We use a unified evaluation prompt that provides the same validity criteria and requires each judge to independently classify a trace as *valid* or *invalid* and report a confidence score in the range [0, 1]. To reduce the effect of stochastic judgment variation, we query each judge three times for each trace, yielding nine judgments per trace.

We aggregate the nine judgments using confidence-weighted voting. For a candidate label ℓ , its normalized support is defined as:

$$S(\ell) = \frac{\sum_{m \in \mathcal{M}} \sum_{r=1}^3 c_{m,r} \cdot \mathbf{1}[y_{m,r} = \ell]}{\sum_{m \in \mathcal{M}} \sum_{r=1}^3 c_{m,r}}. \quad (1)$$

We assign the label with the larger normalized support and use its support as the aggregate confidence. Traces with an aggregate confidence above 0.8 are automatically assigned the corresponding label; the remaining traces are adjudicated independently by two human annotators. The resulting labels define the valid and invalid reasoning-trace groups used in RQ2. The final dataset contains 2,452 valid and 924 invalid traces. R1-0528 contributes 1,439 valid and 251 invalid traces, while R1-Qwen3-8B contributes 1,013 valid and 673 invalid traces.

2.2.3 Reasoning Action Taxonomy and Annotation. Reasoning traces are free-form natural-language text, making systematic comparison across tasks and models difficult. To enable structured analysis, we represent each reasoning trace as a sequence of fine-grained *reasoning actions*, where a reasoning action is defined as a contiguous text span expressing a distinct reasoning behavior.

Drawing on prior studies of LLM reasoning [12, 20, 22, 35], we define a taxonomy of eight reasoning actions that captures the major stages of code reasoning, ranging from task understanding and solution construction to verification and reflection. To strengthen its validity, the taxonomy is further inspected by two researchers with over five years of experience in code intelligence systems, who independently evaluated the coverage, granularity, and ambiguity of the action definitions and iteratively refined them to resolve overlaps. The resulting taxonomy provides a task-independent representation of reasoning behaviors across different coding tasks. Table 1 summarizes the eight reasoning actions.

Using this taxonomy, we automatically annotate each reasoning trace with DeepSeek-V3.2 [7] and Gemini-2.5 Pro [24]. Specifically, each reasoning trace is segmented into contiguous spans,

and each span is assigned exactly one reasoning action. To ensure consistent annotations, we require the models to generate structured JSON outputs containing the assigned reasoning actions, corresponding text spans, and their ordering. The annotation prompt specifies (1) the definitions of the eight reasoning actions with representative examples, (2) segmentation guidelines covering boundary criteria, granularity, and special-case handling, (3) priority rules for resolving ambiguous cases, and (4) the required JSON output schema. The complete annotation prompt is included in our replication package.

To assess annotation reliability, two human annotators independently re-annotated a randomly sampled subset of reasoning traces. Their annotations achieved substantial agreement with the LLM-generated annotations (Cohen’s $\kappa = 0.816$) [5], indicating that the proposed taxonomy and annotation protocol produce reliable reasoning-action annotations.

2.2.4 Action Trace Construction. The annotated reasoning actions are converted into an *action trace*, defined as an ordered sequence of reasoning actions:

$$\tau = (s_1, s_2, \dots, s_n),$$

where each $s_i \in \mathcal{C} = \{PU, SD, IP, CC, VV, KR, MR, AUX\}$ denotes one reasoning action from the taxonomy, forming the canonical representation for all subsequent analyses.

To emphasize transitions between reasoning stages while reducing the influence of repeated elaboration within the same action, we further derive a *collapsed action trace* $\bar{\tau}$ by merging adjacent identical actions. For example,

$$(SD, SD, KR, SD, VV, VV) \rightarrow (SD, KR, SD, VV).$$

2.2.5 Action Trace Measurement. We characterize each reasoning action trace using two groups of metrics: *size metrics*, which quantify the length and output volume of a reasoning trace, and *structural metrics*, which capture the distribution and transition dynamics of reasoning actions.

Size metrics quantify the length of the reasoning trace at both the action and token levels. We report the raw action-trace length n and the collapsed action-trace length m , where consecutive identical actions are merged in the collapsed trace. Their ratio n/m measures the degree of consecutive action repetition. We also report the number of reasoning tokens generated by the model, computed using a consistent tokenizer across all comparisons. These tokenizer-based counts characterize generated reasoning volume and are distinct from API-billed token usage.

Structural metrics characterize the composition of reasoning actions and their transition dynamics. For each action c , its *allocation* is defined as the fraction of action spans in a trace assigned to c . To characterize reasoning flow, we construct a transition matrix P , where P_{ab} denotes the probability of transitioning from action a to action b . We estimate P_{ab} by counting adjacent $a \rightarrow b$ transitions in collapsed traces and normalizing by the total number of outgoing transitions from a . We then quantify transition diversity using the stationary-distribution-weighted average of the row-wise transition entropies:

$$H(P) = \sum_a \pi_a \left(- \sum_b P_{ab} \log_2 P_{ab} \right), \quad (2)$$

where π denotes the stationary distribution of P . Higher entropy indicates greater diversity in the actions that can follow a given action, reflecting a less concentrated transition structure.

2.3 Reasoning Pattern Mining

This section describes how we mine reasoning patterns from the extracted reasoning action traces and derived metrics. We characterize reasoning patterns at two granularities: *macro patterns* at the task level and *micro patterns* at the action level, corresponding to RQ1 and RQ2, respectively.

2.3.1 Reasoning Pattern Definition. A *reasoning pattern* is an action-fragment signature that frequently recurs across a collection of action traces. The support of a signature is the fraction of traces in a trace set whose collapsed action trace contains that signature. A signature is considered a candidate reasoning pattern when its support exceeds a predefined minimum support threshold. We use these patterns as observable regularities in action traces rather than as direct evidence of the model's latent cognitive mechanism.

We represent reasoning patterns using different abstractions across granularities. *Macro patterns* capture high-level reasoning strategies through ordering relations between actions. We denote $A \Rightarrow B$ when action A precedes action B in a trace, allowing arbitrary intermediate actions and thus capturing coarse-grained ordering dependencies.

Micro patterns capture fine-grained reasoning dynamics based on adjacent action transitions. We denote $A \rightarrow B$ as a direct transition between consecutive actions, and $A \leftrightarrow B$ as a short bidirectional loop where two actions alternate. These representations capture local reasoning structures and enable contrastive analysis of effective behaviors and failure-prone anti-patterns.

2.3.2 Mining General Patterns. Macro patterns capture task-level regularities in reasoning behavior. For each task, we extract recurring action-fragment signatures from collapsed action traces. A signature is defined as a frequently occurring action sequence, such as $PU \Rightarrow SD \Rightarrow IP \Rightarrow CC$, indicating that these actions tend to co-occur in this order within traces of the same task.

We consider three classes of signatures: contiguous action n -grams (length 4–5), non-contiguous ordered action paths, and short cyclic loops. A signature is considered task-discriminative if it is frequent within the task and relatively rare in other tasks. The derived metrics interpret these signatures: action allocation indicates which actions dominate a task, transition probabilities describe how it moves between actions, and transition entropy summarizes whether the reasoning process is compact or dispersed.

2.3.3 Mining Micro Patterns. Micro patterns capture fine-grained reasoning dynamics within traces. We analyze valid and invalid reasoning action traces from two perspectives to identify micro patterns and anti-patterns that characterize effective and ineffective reasoning behaviors.

The first perspective concerns *action allocation*. We define the *closure share* as the combined proportion of spans labeled verification and validation (VV), code construction (CC), or auxiliary (AUX). We use this aggregate as a proxy for convergence-oriented allocation; *AUX* retains its taxonomy definition of auxiliary statements and does not exclusively denote answer finalization. The *drift share* is the proportion of spans labeled solution design, meta-reasoning, or knowledge retrieval, capturing behaviors that revisit or expand the reasoning space. We compute the *closure-minus-drift* score by subtracting drift share from closure share; positive values indicate more convergence-oriented allocation, whereas negative values indicate more drift-oriented allocation.

The second perspective focuses on *local structural patterns*. We extract contiguous action n -grams from collapsed action traces and represent them as local transitions $A \rightarrow B$ and short loops $A \leftrightarrow B$. These patterns capture fine-grained procedural regularities that are not visible at the aggregate allocation level: allocation metrics characterize whether reasoning tends toward convergence (closure) or exploration (drift), whereas local patterns reveal the concrete action-level fragments through which such tendencies are realized.

We first apply an effect-size screen: a candidate pattern must exceed a minimum support threshold in its enriched group and show a minimum support contrast against the comparison group. Patterns that are more frequent in invalid traces than in valid ones are treated as anti-patterns. We then audit the retained contrasts using two-sided Fisher’s exact tests with Benjamini–Hochberg correction [2].

2.4 Leveraging Reasoning Patterns

This section describes how we leverage the discovered reasoning patterns to improve AI coding performance. We evaluate their effectiveness in three representative scenarios for RQ3:

Coding Task Classification We train a supervised probe on reasoning-pattern features extracted from action traces to predict the coding task performed by an LRM. These features operationalize the macro patterns identified in RQ1 by encoding how a trace allocates actions, transitions between them, and distributes them across reasoning stages.

In total, we construct 57 features organized into seven groups: (1) eight action-allocation features; (2) two features capturing trace length and compression; (3) twelve key transition-probability features; (4) three repetition features capturing self-repetition and *VV*- or *SD*-centered iterative behaviors; (5) twenty-four position-aware action-allocation features that model where actions occur in the trace; (6) four action-effort features; and (7) four aggregate features measuring verification density, design ratio, reflection ratio, and code-output ratio. The probe is restricted to reasoning-pattern features and does not access task descriptions, final answers, or any external metadata.

Pattern-Guided Prompting We incorporate the micro patterns identified in RQ2 into task prompts, integrating positive patterns from valid traces with negative constraints from anti-patterns. Positive patterns encourage transitions toward implementation, verification, and closure, while anti-pattern constraints discourage undesirable loops such as $SD \rightarrow KR \rightarrow SD$ and $VV \rightarrow SD \rightarrow MR$, steering the model toward structured, goal-directed reasoning.

Early Stopping Negative Patterns The online monitor uses a local rule-based action mapper, distinct from the full-trace LLM annotation used to discover patterns in RQ1 and RQ2. It buffers streamed reasoning_content until punctuation or a newline completes a micro-unit, then assigns that unit one of the eight fine-grained actions and an Explore, Build, Verify, or Wrap state using lexical cues. A coarse-state change starts a new segment; a fine-action change does so when accompanied by a transition cue or after at least three units, while high lexical novelty requires a transition cue. Each closed segment takes the majority label of its units. Adjacent identical segment labels are collapsed before matching task-specific motifs. This mapper makes no separate annotation-model or API call.

At each closed segment, the policy combines failure motifs (e.g., $SD \rightarrow KR \rightarrow SD$ and $VV \rightarrow MR \rightarrow SD$) and healthy motifs (e.g., $VV \rightarrow IP \rightarrow VV$ and $IP \rightarrow VV \rightarrow AUX$) with recent exploration, novelty, and build/verify history. A loop trigger requires sustained high drift risk and low closure evidence subject to task-specific minimum-progress guards; a separate closure trigger requires a high closure score together with answer or wrap evidence. A stream-level repeated-text check can also stop de-generate output before segment closure. When a trigger fires, the runner closes the initial stream and sends the original task and stopped reasoning prefix in a second request to the same target model for a final answer. This request asks for answer-only output and disables thinking where supported; any reasoning actually emitted by the finalizer remains part of its generated output.

Table 2. Macro Reasoning Patterns across Four Coding Tasks

Task	Macro Pattern	Representative Signature	Support (within-task vs. cross-task)	
			R1-0528	R1-Qwen3-8B
Generation	Design-to-implementation	$PU \Rightarrow SD \Rightarrow IP \Rightarrow VV$	91.4% vs. 17.0%	88.0% vs. 22.0%
Execution Reasoning	State tracing	$PU \Rightarrow \{VV, MR\}^*$	45.9% vs. 2.2%	53.5% vs. 3.4%
Debugging	Evidence-fix-recheck	$\{SD, IP\} \Rightarrow CC \Rightarrow VV$	68.1% vs. 21.5%	70.8% vs. 20.4%
Translation	Semantic mapping-to-implementation	$PU \Rightarrow SD \Rightarrow IP \Rightarrow CC$	36.5% vs. 9.5%	30.1% vs. 16.2%

3 RQ1: General Reasoning Patterns

3.1 Setup

RQ1 characterizes task-level reasoning behaviors using the macro pattern mining procedure described in Section 2.3.2. We analyze 3,376 collapsed reasoning action traces generated by two LRMs across four coding tasks. A macro pattern is retained if its within-task support exceeds a predefined threshold of 0.05 and its support difference from the other tasks exceeds a margin of 0.03. These hyperparameters are calibrated by two experts based on empirical inspection of preliminary results.

3.2 Results

Across the four coding tasks, we identify a distinct macro reasoning pattern for each task. Table 2 reports one representative macro signature per task and its within- versus cross-task support. The large support gaps (e.g., 91.4% vs. 17.0% for code generation and 45.9% vs. 2.2% for code execution reasoning on R1-0528) indicate that these signatures capture task-specific strategies rather than generic reasoning behaviors.

3.2.1 Code Generation Follows a Design-to-Implementation Pattern. Both models organize code generation as progressive refinement from high-level solution design to implementation planning and verification. R1-0528 interleaves design, planning, and verification ($PU \rightarrow SD \rightarrow VV \rightarrow SD \rightarrow IP \rightarrow VV$), with $IP \leftrightarrow VV$ as the dominant local interaction. R1-Qwen3-8B follows the same overall structure but adds meta-reasoning ($PU \rightarrow SD \rightarrow MR \rightarrow SD \rightarrow VV$).

3.2.2 Code Execution Reasoning Follows a State-Tracing Pattern. Both models perform code execution reasoning by progressively simulating program states and validating intermediate results. R1-0528 follows a compact state-tracing and output-verification trajectory ($PU \rightarrow VV \rightarrow MR$) with the lowest transition entropy among all tasks (1.690). R1-Qwen3-8B follows the same overall strategy through longer verification-reflection cycles ($PU \rightarrow VV \rightarrow MR \rightarrow VV$), resulting in a more iterative process.

3.2.3 Program Debugging Follows an Evidence-Fix-Recheck Pattern. Both models organize debugging as an iterative cycle of fault validation, code modification, and re-verification. R1-0528 enters verification early and repeatedly alternates between IP and VV ($PU \rightarrow VV \rightarrow IP \rightarrow VV \rightarrow CC \rightarrow VV$), supported by frequent $PU \rightarrow VV$ and $CC \rightarrow VV$ transitions. R1-Qwen3-8B follows the same pattern but more often reopens SD after verification ($PU \rightarrow SD \rightarrow VV \rightarrow SD \rightarrow IP \rightarrow CC \rightarrow VV \rightarrow CC$), suggesting a more exploratory debugging strategy.

3.2.4 Code Translation Follows a Semantic Mapping Pattern. Both models begin by constructing semantic correspondences between the source and target programs before proceeding to implementation. R1-0528 transitions more directly from semantic mapping to implementation and validation ($PU \rightarrow SD \rightarrow IP \rightarrow VV \rightarrow IP$), whereas R1-Qwen3-8B revisits SD more frequently

through $PU \rightarrow SD \rightarrow IP \rightarrow SD \rightarrow IP$. The shared signature $PU \Rightarrow SD \Rightarrow IP \Rightarrow CC$ indicates that code translation is dominated by semantic reconstruction followed by code construction, rather than simple textual transformation.

3.2.5 Shared Reasoning Scaffold. Despite their distinct objectives, all four coding tasks exhibit a common understand-then-act reasoning scaffold. PU serves as the entry point in more than 95% of reasoning traces across both models, after which reasoning diverges into task-specific sequences involving SD , IP , VV , and CC . VV remains central in the representative signatures of generation, execution reasoning, and debugging, while IP is especially prominent in generation and translation. The raw-to-collapsed length ratio remains relatively stable across tasks (1.3–1.7), indicating moderate local repetition while preserving similar high-level reasoning structures. Overall, task-specific behaviors emerge only after the shared problem-understanding stage.

3.2.6 Cross-Model Comparison. The two [primary R1](#) models exhibit highly consistent macro reasoning patterns while differing in how these patterns are realized. Across all four tasks, R1-Qwen3-8B generates longer reasoning traces than R1-0528 (+37.1% for code generation, +133.1% for code execution reasoning, +21.8% for program debugging, and +7.6% for code translation) and consistently shows equal or higher transition entropy, particularly for execution reasoning and debugging. These results indicate that both models adopt the same task-level reasoning strategies, but R1-Qwen3-8B realizes them through more iterative and dispersed trajectories, whereas R1-0528 follows more compact paths.

Finding 1. Across the two R1 models, each coding task exhibits a distinct macro reasoning pattern that reflects its underlying problem-solving strategy. Code generation follows a design-to-implementation pattern, code execution reasoning relies on state tracing, program debugging follows an evidence-fix-recheck cycle, and code translation is driven by semantic mapping followed by implementation.

4 RQ2: Micro Reasoning Patterns

4.1 Setup

RQ2 characterizes fine-grained reasoning behaviors by comparing valid and invalid reasoning action traces using the micro pattern analysis procedure described in Section 2.3.3. We analyze 2,452 valid traces and 924 invalid traces generated by [the two primary R1 models](#) across four coding tasks. A micro pattern is retained if its enriched-group support exceeds 0.08 and its cross-group support difference exceeds 0.06. These thresholds are calibrated by two domain experts based on empirical inspection of preliminary results.

4.2 Results

We first examine differences between valid and invalid reasoning action traces, and then present the resulting micro patterns and anti-patterns that characterize effective and ineffective reasoning trajectories. These analyses reveal differences in fine-grained action dynamics between valid and invalid reasoning traces.

4.2.1 Invalid Traces Shift Action Allocation Toward Drift-oriented Actions. Table 3 reports reasoning-token counts, action lengths, and allocation statistics for valid and invalid traces, aggregated over the four oracle-backed tasks. Across both models, invalid traces are consistently longer than valid traces in tokens, raw action length, and collapsed action length, showing that traces judged invalid contain more generated reasoning rather than fewer reasoning steps.

Table 3. Length and Allocation Statistics for Valid and Invalid Reasoning Traces

Metric	R1-0528		R1-Qwen3-8B	
	Valid	Invalid	Valid	Invalid
Reasoning tokens	3,690	6,609	4,330	6,991
Raw action length	18.58	24.96	22.64	32.21
Collapsed action length	11.78	14.91	15.97	21.99
Closure share	0.414	0.241	0.330	0.219
Drift share	0.247	0.391	0.313	0.464
Closure–Drift	+0.167	−0.150	+0.017	−0.245

Table 4. Identified Micro Reasoning Patterns and Anti-Patterns

Task	Model	Micro Pattern	Micro Anti-pattern
Generation	R1-0528	IP→VV→AUX	KR→VV→MR
		IP→VV→IP→VV→IP	SD→KR→IP
	R1-Qwen3-8B	VV→IP→VV	KR↔SD
		VV→IP→VV→IP	MR↔KR
Execution	R1-0528	PU→VV→MR→VV	MR→PU→MR
	R1-Qwen3-8B	PU→VV→MR→VV	MR→SD→{PU,VV}
Debugging	R1-0528	CC→VV→AUX	VV→SD→MR
			SD→MR→IP
	R1-Qwen3-8B	VV→IP→VV	SD→KR→SD
Translation	R1-0528	SD→IP→CC	SD→MR→SD→MR→SD
			VV→IP→CC
	R1-Qwen3-8B	PU→SD→IP→MR	SD→MR→SD→IP PU→SD→PU→SD

Beyond these length differences, valid and invalid traces also differ in how reasoning actions are distributed. Invalid traces systematically shift allocation from closure-oriented to drift-oriented actions: the closure share decreases substantially (R1-0528: 0.414 → 0.241; R1-Qwen3-8B: 0.330 → 0.219), while the drift share increases (R1-0528: 0.247 → 0.391; R1-Qwen3-8B: 0.313 → 0.464), reflecting a shift from verification, code construction, and auxiliary actions toward solution design, meta-reasoning, and knowledge retrieval.

At the level of aggregate structure, this effect is captured by the closure-minus-drift score, which reverses sign between valid and invalid traces (R1-0528: +0.167 → −0.150; R1-Qwen3-8B: +0.017 → −0.245), indicating that invalid reasoning is dominated by exploratory and revisiting behaviors rather than merely lacking verification.

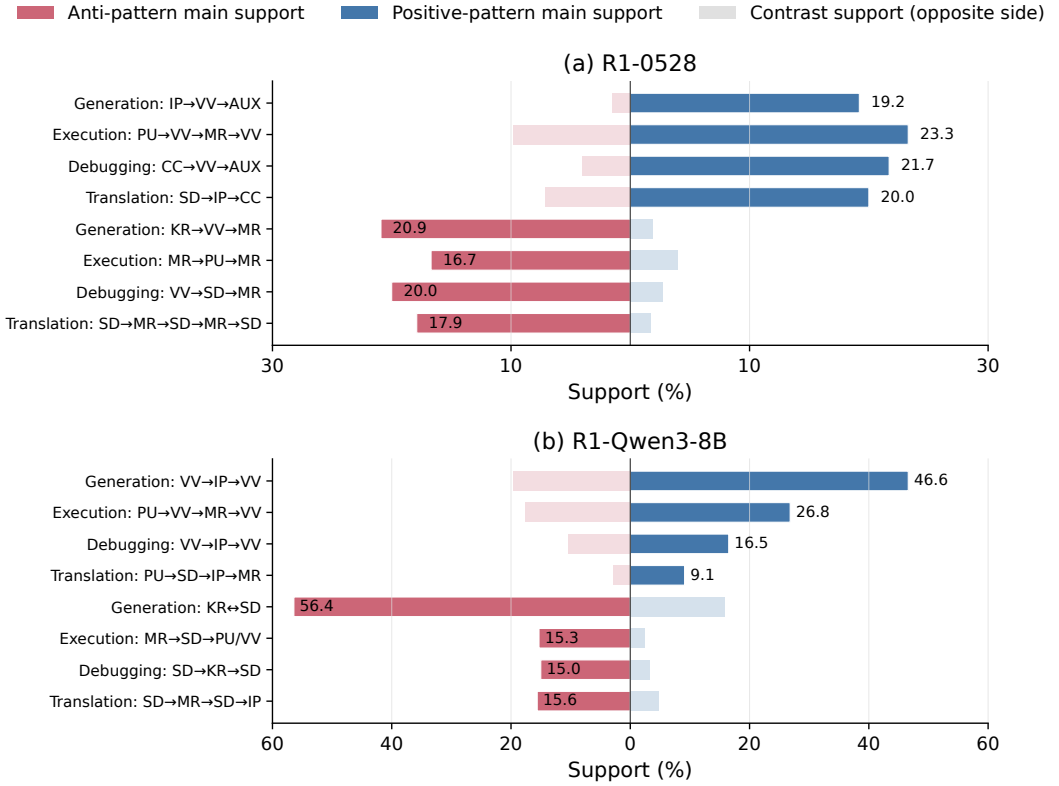


Fig. 2. Support Comparison of Micro Reasoning Patterns and Anti-patterns.

4.2.2 Micro Patterns Distinguish Productive Iteration from Reopening. We next examine whether these shifts are reflected in fine-grained local structures. Table 4 summarizes the discovered micro patterns and anti-patterns, while Figure 2 visualizes support contrasts of representative high-support patterns across task-model settings. The results show that these differences are also manifested at the micro-pattern level. For instance, in R1-Qwen3-8B on code generation, the positive pattern $VV \rightarrow IP \rightarrow VV$ appears in 46.6% of valid traces but only 19.6% of invalid traces. In contrast, the anti-pattern $KR \leftrightarrow SD$ appears in 56.4% of invalid traces but only 15.9% of valid traces. Similarly, for R1-0528 on debugging, the pattern $CC \rightarrow VV \rightarrow AUX$ is substantially more frequent in valid traces (21.7% vs. 4.0%), whereas $VV \rightarrow SD \rightarrow MR$ is strongly associated with invalid traces (20.0% vs. 2.7%).

These patterns reveal a clear distinction in iterative behavior. Productive iteration typically follows a verify-adjust-reverify or verify-commit structure, consistent with prior findings that execution and self-verification improve reliability in code reasoning [13]. In contrast, invalid traces exhibit repetitive transitions centered on design revision (SD), reflection (MR), or knowledge recall (KR), without progressing toward implementation or closure.

We further check representative support contrasts using two-sided Fisher’s exact tests with Benjamini–Hochberg correction, and report the complete support, count, p -value, and q -value evidence in the replication package. These results suggest that the association between iteration and trace validity depends on the actions involved and the direction of their transitions.

4.2.3 Cross-model Consistency with Task-dependent Strength. Finally, we examine the generality of the observed validity contrast across models and tasks. The closure-versus-reopening distinction is consistent across [the two primary R1 models, indicating that the contrast is not limited to one model scale within that lineage.](#)

However, the strength of this contrast varies across settings. It is more pronounced in R1-Qwen3-8B, where invalid traces exhibit stronger cycles of design revision, reflection, and knowledge recall, consistent with persistent non-convergent exploration in traces judged invalid.

Across tasks, the effect is strongest in code generation, where invalid traces are substantially longer (17,163 vs. 7,102 tokens for R1-0528; 14,925 vs. 8,598 for R1-Qwen3-8B) and heavily concentrated in drift-oriented behaviors. Code execution and debugging tasks exhibit similar but more structured patterns, while translation shows weaker and less symmetric differences, making it a less definitive case for behavioral separation.

Finding 2. Valid and invalid reasoning traces differ not only in length but also in their action allocation and micro patterns, with valid traces showing verification-driven convergence and invalid traces exhibiting anti-patterns of repeated design, reflection, and knowledge recall without closure.

5 RQ3: Applications of Reasoning Patterns

To assess the practical utility of the discovered reasoning patterns for efficient AI coding, we conduct a set of targeted studies across code-related tasks.

- **RQ3.1** *Are reasoning patterns discriminative enough to distinguish different AI coding tasks? (Section 5.1)*
- **RQ3.2** *How does pattern-guided prompting affect task performance and token usage? (Section 5.2)*
- **RQ3.3** *What accuracy-token trade-off does pattern-aware early stopping produce? (Section 5.3)*

5.1 Task Classification from Macro-pattern Features

Setup. Following the task recovery probe defined in Section 2.4, RQ3.1 evaluates whether task identity can be inferred from reasoning-pattern features alone. The evaluation dataset consists of 3,376 reasoning traces generated by two models across four coding tasks. We evaluate two standard classifiers, logistic regression and random forest, using 5-fold stratified cross-validation. Performance is measured in terms of classification accuracy, reported as mean and standard deviation across folds; the random baseline is 25% for the four-class setting.

Results. Figure 3 reports task classification performance using only reasoning-pattern features derived from action traces. Error bars indicate cross-validation standard deviation, and the dashed line denotes the random baseline (25%).

Using 57 reasoning-pattern features, logistic regression achieves 0.861 ± 0.032 accuracy for R1-0528 and 0.770 ± 0.019 for R1-Qwen3-8B. Random forest yields comparable results (0.864 ± 0.027 and 0.775 ± 0.022 , respectively), [showing task separability for both models within the R1 lineage and across the two classifiers.](#) Since the models operate solely on reasoning-pattern features without access to task descriptions or outputs, these results demonstrate that reasoning traces preserve task identity in their structural pattern organization.

To understand which aspects of reasoning patterns drive this separability, we examine feature importance based on mutual information. For both models, the most informative features consistently relate to how reasoning effort is distributed across core actions (e.g., design ratio, and the

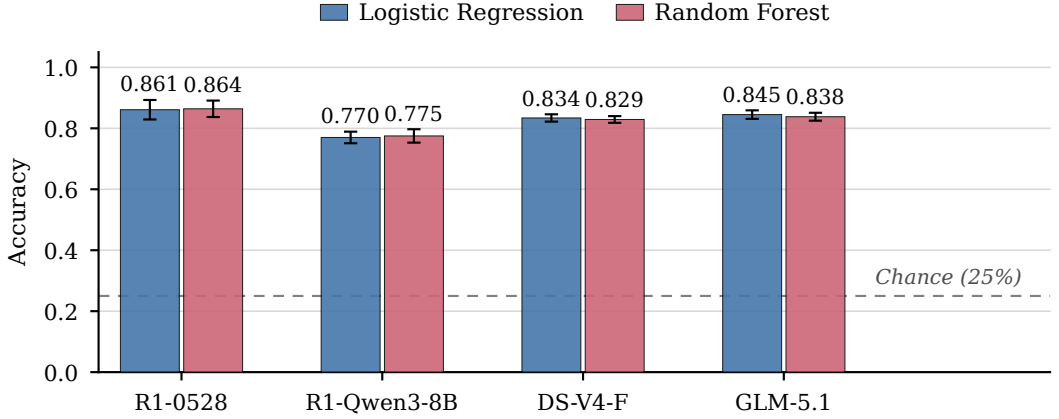


Fig. 3. Task Identification from Reasoning Pattern Features.

distributions of *PU*, *SD*, *IP*, and *VV*). Overall, the feature-importance analysis highlights action allocation as an informative signal for task identification.

Finding 3.1. Reasoning-pattern features alone are sufficient to identify coding tasks with high accuracy, despite not using task content, answers, or metadata, suggesting that action traces inherently encode task-discriminative reasoning structures.

5.2 Micro-Pattern Guided Prompting

Setup. RQ3.2 explores whether micro reasoning patterns can be operationalized as inference-time guidance to improve coding performance. We compare two prompting strategies: a *default prompt*, which uses the original task instruction without additional guidance, and a *pattern-guided prompt*, which augments the same instruction with task-specific guidance derived from mined micro patterns.

We evaluate direct guidance derived from each primary R1 model's own patterns on four coding tasks with objective correctness oracles. To test cross-family transfer, we also apply the R1-family pattern inventories to DeepSeek-V4-Flash (DS-V4-F) [8] and GLM-5.1 [37].

Performance is measured using pass@1 [3]. To reduce variance from stochastic decoding, each setting is run three times, and we report mean pass@1 as the average across independent runs. We additionally report average per-instance generated-output token counts (in thousands) to characterize output length.

Results. Table 5 summarizes the effectiveness of micro-pattern guided prompting across four AI coding tasks. We evaluate two settings: *Direct*, which applies patterns mined from the same model, and *Transferred*, which applies guidance drawn from the R1-family pattern inventories to DS-V4-F and GLM-5.1. Neither target model's trace-derived patterns are used to construct these prompts.

1) *Direct Results.* Pattern-guided prompting improves mean pass@1 on all four tasks for both R1-0528 and R1-Qwen3-8B, with the largest gains on debugging for R1-0528 and on execution reasoning and translation for R1-Qwen3-8B(+5.80, +5.50, and +5.05 points, respectively).

Improvement and token usage are task-dependent. For example, R1-0528 debugging improves by 5.80 percentage points while generated-output tokens increase from 4.5K to 6.6K; R1-Qwen3-8B

Table 5. Performance and Cross-Model Transfer of Pattern-Guided Prompting

Task	Method	Direct				Transferred			
		R1-0528		R1-Qwen3-8B		DS-V4-F		GLM-5.1	
		P@1↑	Tok.↓	P@1↑	Tok.↓	P@1↑	Tok.↓	P@1↑	Tok.↓
Generation	Default	72.46	14.9	56.21	17.9	80.78	13.0	90.29	16.5
	Pattern-only	73.14	15.2	55.86	18.3	84.36	13.5	90.06	13.2
	Anti-pattern-only	72.08	14.6	56.39	17.6	79.84	12.5	90.74	13.0
	Pattern+Anti-pattern	74.49	15.6	56.88	17.3	87.80	12.2	91.17	12.9
Execution Reasoning	Default	55.76	1.7	44.24	3.4	56.81	2.1	59.84	1.6
	Pattern-only	57.16	2.4	48.17	4.4	58.21	2.6	60.32	1.6
	Anti-pattern-only	56.63	2.1	46.86	3.9	57.54	2.3	60.41	1.5
	Pattern+Anti-pattern	58.37	2.9	49.74	4.7	59.27	2.8	60.73	1.5
Debugging	Default	88.80	4.5	61.20	7.7	91.40	2.4	93.00	2.8
	Pattern-only	91.20	5.8	61.54	7.8	90.92	2.5	93.02	2.6
	Anti-pattern-only	90.47	5.1	61.08	7.6	92.18	2.2	93.11	2.6
	Pattern+Anti-pattern	94.60	6.6	61.72	7.9	92.80	2.3	93.20	2.5
Translation	Default	55.08	5.7	39.09	4.2	55.05	2.1	56.97	2.5
	Pattern-only	56.81	6.5	42.51	4.4	54.73	2.2	58.03	2.4
	Anti-pattern-only	57.44	6.2	43.27	4.3	56.73	1.8	58.52	2.4
	Pattern+Anti-pattern	58.78	7.3	44.14	4.5	57.44	1.9	58.75	2.3

generation improves by 0.67 points while tokens decrease from 17.9K to 17.3K. These output-token counts exclude the additional prompt tokens.

2) *Ablation: Positive Patterns vs. Anti-pattern Constraints.* To isolate the contribution of each guidance component, we further evaluate two ablated variants: *Pattern-only*, which retains only the positive patterns mined from valid traces, and *Anti-pattern-only*, which retains only the negative constraints derived from anti-patterns. Across the sixteen task–model settings, each component alone yields inconsistent gains and can even underperform the default prompt: Pattern-only degrades R1-Qwen3-8B generation (55.86 vs. 56.21) and DS-V4-F translation (54.73 vs. 55.05), while Anti-pattern-only degrades DS-V4-F generation (79.84 vs. 80.78) and R1-Qwen3-8B debugging (61.08 vs. 61.20). In contrast, their combination achieves the highest pass@1 among all guided variants in every task–model setting and surpasses the default prompt in all sixteen settings. The observed combination effect is consistent with complementary roles for positive patterns and anti-pattern constraints: one may provide a direction toward closure, while the other discourages reopening loops. The current ablations do not establish this mechanism or distinguish pattern-specific effects from generic advice.

3) *Transferred Results.* We next assess the transfer of R1-family guidance to the two additional target models. On DS-V4-F, pass@1 improves on all four tasks, with the largest gain on code generation (+7.02 points); token use increases on execution reasoning and decreases on the other three tasks. On GLM-5.1, pass@1 improves on all four tasks by 0.20–1.78 percentage points, while token use decreases on all four tasks. For generation, pass@1 increases from 90.29 to 91.17 (+0.88 points) and token use falls from 16.5K to 12.9K. Thus, guidance improves mean pass@1 across both transfer models, although the magnitude of improvement and token usage vary by setting.

Table 6. Pattern-aware Early Stopping Performance across Models. Tok.: API-reported token usage (thousands); stopped runs include finalizer output.

Task	Model	Full Reasoning		Pattern-aware Stop			
		P@1↑	Tok.↓	P@1↑	Tok.↓	Δ P@1 (pp)	Δ Tok. (%)
Generation	R1-Qwen3-8B	56.21	17.89K	52.14	10.64K	-4.07	-40.56%
	DS-V4-F	80.78	13.00K	73.36	8.43K	-7.42	-35.13%
	GLM-5.1	90.29	16.54K	88.06	7.85K	-2.23	-52.52%
Execution Reasoning	R1-Qwen3-8B	44.24	3.43K	40.58	2.10K	-3.66	-38.75%
	DS-V4-F	56.81	2.13K	51.57	0.85K	-5.24	-59.78%
	GLM-5.1	59.84	1.64K	58.63	0.52K	-1.21	-68.53%
Debugging	R1-Qwen3-8B	61.20	7.65K	54.40	4.18K	-6.80	-45.32%
	DS-V4-F	91.40	2.42K	86.60	1.69K	-4.80	-30.16%
	GLM-5.1	93.00	2.77K	91.20	1.74K	-1.80	-37.16%
Translation	R1-Qwen3-8B	39.09	4.18K	37.64	2.33K	-1.45	-44.22%
	DS-V4-F	55.05	2.14K	50.79	1.23K	-4.26	-42.65%
	GLM-5.1	56.97	2.53K	55.48	1.51K	-1.49	-40.56%

Finding 3.2. Micro-pattern-guided prompts combining positive patterns and anti-pattern constraints improve mean pass@1 in all 16 task-model settings, including transfer to two additional models. The combined guidance achieves higher mean pass@1 than either component alone in all evaluated settings, suggesting complementary benefits from the two components.

5.3 Pattern-aware Early Stopping

Setup. RQ3.3 examines whether anti-pattern signals can be leveraged for online early stopping to change the accuracy-token trade-off. We compare two settings: *Full Reasoning*, which allows the model to complete its full reasoning process before producing an answer, and *Pattern-aware Stop*, an online stopping policy defined in Section 2.4 that monitors the streaming reasoning trace and triggers answer finalization when a drift, closure, or repeated-text condition is met. Since Pattern-aware Stop requires access to streaming reasoning outputs, we apply it to R1-Qwen3-8B and two additional target models, DS-V4-F and GLM-5.1, while R1-0528 is excluded due to the lack of a streaming-capable interface during evaluation. Following RQ3.2, each setting is executed three times, and we report mean pass@1 together with average per-instance API-reported token usage.

The Tok. metric is taken from the same target-model API usage field for both settings and includes finalizer output for stopped runs. The local rule-based monitor makes no separate annotation-model call, whereas the second request to the target model resends the task and stopped reasoning prefix. Tok. measures API-reported token usage rather than elapsed time or monetary cost.

In an offline replay of 40 stored R1-Qwen3-8B traces split into 9,019 25-character chunks, local monitor processing took 0.13 ms per chunk at the median (0.30 ms at the 95th percentile); cumulative local CPU time was 19 ms per trace at the median (142 ms at the 95th percentile). This microbenchmark excludes model, network, and finalizer latency.

Results. Table 6 reports the RQ3.3 comparison across four coding tasks. For R1-Qwen3-8B, reported token usage decreases by 38.75%–45.32%, while pass@1 decreases by 1.45–6.80 percentage points. Translation has the smallest accuracy decrease (1.45 points) at a 44.22% token reduction; generation and debugging have larger accuracy losses.

Reported token usage also decreases on the two additional models, DS-V4-F (30.16%–59.78%) and GLM-5.1 (37.16%–68.53%). Their pass@1 decreases by 4.26–7.42 and 1.21–2.23 points, respectively. These results show an accuracy–token trade-off across the evaluated models.

Finding 3.3. Pattern-aware early stopping reduces API-reported token usage by 30–69%, with a 1.21–7.42 percentage-point decrease in pass@1 across the three evaluated models, demonstrating an accuracy–token trade-off across task–model settings.

6 Discussion

6.1 Understanding Reasoning Structures

Our results suggest that reasoning in AI coding is organized around recurrent action-level structures rather than isolated step sequences or linear chains of thought. Across tasks, we observe stable macro-level behavioral regularities, indicating that reasoning traces exhibit compositional structure that is consistent within tasks but distinct across them.

At a finer granularity, the contrast between valid and invalid traces suggests that reasoning quality is closely related to how actions evolve toward convergence. This is reflected in the identified micro-level patterns and anti-patterns, which capture differences in action-level progression between traces judged valid and invalid.

6.2 Future Research Directions

Our findings suggest several promising directions for future research on reasoning in AI coding.

6.2.1 Evaluation of Reasoning Models. Reasoning traces provide structured behavioral signals that are not captured by outcome-based metrics such as pass@1 or trace length. This motivates future work on evaluation frameworks grounded in action-level reasoning structure, enabling finer-grained assessment of model reasoning behavior.

6.2.2 Structure-Aware Inference Control. The discovered reasoning patterns suggest the potential for controlling inference through structural signals, enabling fine-grained monitoring and intervention during reasoning, including but not limited to early stopping.

6.2.3 Learning with Reasoning Structure. Beyond pattern-guided prompting, pattern-augmented reasoning traces expose intermediate structures that may serve as supervision signals for training models that explicitly optimize reasoning processes.

7 Threats to Validity

Construct Validity. Our analysis abstracts free-form reasoning traces into discrete reasoning actions, enabling systematic comparison but potentially losing fine-grained semantic distinctions and internal reasoning nuances. We mitigate this limitation by grounding the taxonomy in prior work, refining it via expert inspection, and validating automatic annotations against human re-labeling (Cohen’s $\kappa = 0.816$). However, alternative taxonomies, segmentation granularities, or collapsing rules may lead to different patterns, suggesting the need for sensitivity analysis in future work.

Internal Validity. Trace-validity labels are derived from an LLM-based judge ensemble, with human adjudication of low-confidence cases. Confidence-weighted voting reduces judgment variance, but these labels may still miss subtle reasoning errors. Task-level functional correctness is evaluated separately using objective oracles and summarized as pass@1. Nonetheless, prompting and early-stopping results should be interpreted as intervention evidence rather than causal attribution to specific mined patterns, motivating future controlled interventions for causal validation.

For early stopping, the online rule-based mapper is distinct from the offline LLM action annotators, and their agreement on partial prefixes has not yet been measured. Mapping errors may therefore alter motif detection and stopping decisions. The stopping policy also includes a repeated-text trigger independent of action motifs.

External Validity. Our study spans four coding tasks and four model configurations, which may not fully represent the diversity of software engineering scenarios or model architectures. R1-0528 and its Qwen3-8B distillation share a lineage, and DS-V4-F is another DeepSeek model; GLM-5.1 provides one external model-family test. Further validation across tasks, languages, and model families is required to assess generality. GLM-5.1 is an additional target model in RQ3; it also participated in the validity-judge ensemble for the primary R1 traces. Its intervention results thus provide cross-family evidence, but they are not a fully independent replication of the entire discovery-to-intervention pipeline.

8 Related Work

Recent LRMs incorporate extended reasoning traces as a core component of inference [6]. In code-related settings, prior work shows that reasoning-before-coding is not uniformly beneficial and depends on how reasoning is structured relative to code generation [18]. Other studies explore leveraging external or structured signals to improve reasoning, such as mining discussion data into training chains [34] or validating intermediate steps via search-based refinement [28]. These efforts demonstrate the utility of explicit reasoning for code generation, but primarily focus on improving outputs or training signals rather than characterizing recurring reasoning structures across tasks.

Prior work also examines reasoning quality and failure modes. Studies show that reasoning quality is not reliably aligned with final correctness, as flawed reasoning may still produce correct outputs and vice versa [39], and that superficial signals such as step count or verbosity are weak indicators of correctness [31]. Recent work further categorizes reasoning behaviors and failure patterns such as hallucination, overthinking, and erroneous backtracking [1, 12, 23]. While these studies move beyond outcome-only evaluation, they are often limited to single tasks or rely on coarse-grained or manual annotations, leaving open how reasoning structures vary across tasks and how they differ between valid and invalid traces.

Reasoning traces have also been studied for inference-time control. Methods such as SEER adapt reasoning strategies during generation [10], while work on overthinking motivates compressing redundant reasoning steps [4, 14, 38]. Early-stopping approaches exploit intermediate signals to terminate generation based on convergence or confidence criteria [19, 33]. These methods primarily optimize efficiency or decoding behavior, but do not explicitly derive reasoning patterns from empirical trace analysis.

In contrast, we treat reasoning traces as empirical behavioral artifacts, abstracting them into reasoning action traces and analyzing recurring structures across code generation, code execution reasoning, program debugging, and code translation. This enables a unified characterization of task-level patterns and action-level (anti-)patterns, and further connects these structures to downstream applications in task identification, pattern-guided prompting, and pattern-aware early stopping.

9 Conclusion

In this paper, we present an empirical study of reasoning patterns in AI coding. By abstracting reasoning traces into action sequences, we identify macro-level patterns that characterize task-specific reasoning strategies and micro-level patterns associated with valid and invalid reasoning traces across four coding tasks. We further show that these patterns are actionable, enabling task

identification, improved prompting, and **reduced API-reported token usage through early stopping, accompanied by a measured pass@1 trade-off.**

Data Availability

All code and data used in this study are publicly available at: https://anonymous.4open.science/r/FSE_2027. The repository includes: (i) the collected reasoning traces and annotated datasets across the four tasks (code generation, code execution reasoning, debugging, and translation); (ii) the code for pattern discovery, pattern-guided prompting and pattern-aware early stopping (RQ1–RQ3); and (iii) all experimental results and evaluation scripts.

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